# George Zhang

(778) 682-9123 | gzhang06@outlook.com | linkedin.com/in/iamgeorgezhang | github.com/TheYellowDuck | georgezhang.ca

Canadian & Hong Kong citizen · eligible for U.S. internships (J-1); TN-eligible post-graduation

## Education

**University of Waterloo** Waterloo, ON

*Bachelor of Computer Science (Honours, Co-op)* *Sep 2024 – May 2029*

- Relevant coursework: Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Optimization, Logic & Computation, OOP (C++).

## Experience

**Ford Motor Company** Waterloo, ON

*Software Developer Intern — Android Automotive OS* *May 2026 – Present*

- Develop and ship features for the in-vehicle Car Dialer app on Android Automotive OS — Ford's production infotainment platform — across a multi-module Java/Kotlin codebase (Android Telecom, Bluetooth HFP), with Robolectric unit tests and Gradle / AOSP-Soong CI.

**University of British Columbia** Vancouver, BC

*Undergraduate Research Assistant — Centre for Brain Health* *Jun 2025 – Aug 2025*

- Led development of two full-stack hand-rehabilitation apps within PIKA, an AI Parkinson's-care platform deployed at Vancouver Coastal Health.
- Built a real-time computer-vision pipeline (Python, OpenCV, MediaPipe) detecting 7 hand-exercise gestures at ∼90% accuracy with <100 ms latency, fully on-device to meet PIPEDA/GDPR requirements.
- Designed activity-heatmap dashboards that cut clinician monitoring overhead by 30%; presented at the 11th Singapore International Parkinson Disease & Movement Disorders Symposium.

## Projects

**Code-Aware RAG System** | Python, Transformers github.com/TheYellowDuck/RAG-codebase

- Engineered a retrieval-augmented-generation system answering codebase questions with citation-backed answers — AST-boundary chunking (tree-sitter), dense + BM25 fusion (RRF), a code graph with personalized PageRank, a validated listwise LLM reranker, and a RAGAS-style faithfulness check; provider-agnostic (Anthropic/OpenAI/local).
- Validated every choice as a measured ablation (recall@k, MRR, NDCG, bootstrap CIs, paired significance): the LLM reranker significantly improved ranking (MRR/NDCG, $p<0.001$) while a cross-encoder proved net-negative on code.

**RLHF Pipeline** | PyTorch github.com/TheYellowDuck/rlhf-pipeline

- Reproduced the LLM post-training stack (SFT, reward modeling, PPO) with the RL core written from scratch: GAE, the clipped policy-gradient surrogate, per-token KL-to-reference shaping, and an adaptive KL controller.
- Added DPO and GRPO alternatives, LoRA parameter-efficient fine-tuning, multi-GPU training (HuggingFace Accelerate), and an independent LLM-as-judge win-rate evaluation with position-bias control.

**Autonomous LLM Web Agent** | Python, Playwright github.com/TheYellowDuck/web-agent

- Designed a ReAct + reflection agent (browser accessibility tree, provider-agnostic across Claude/GPT/Gemini) with a WebArena / Online-Mind2Web eval harness; showed the same agent scores 0.43 vs 0.61 by scorer alone, isolating measurement from real capability.

## Technical Skills

| | |
|---|---|
| **Languages** | Python, C++, Java, Kotlin, TypeScript/JavaScript, SQL |
| **ML / AI** | PyTorch, Transformers, RAG, RLHF (PPO/DPO/GRPO), LLM agents, prompt engineering, LLM-as-judge eval |
| **Data** | NumPy, Pandas, scikit-learn, Matplotlib, OpenCV, MediaPipe, embeddings, statistics |
| **Tools** | Git, Linux, Docker, GitHub Actions, HuggingFace, Playwright, Anthropic/OpenAI APIs |
| **Concepts** | Data Structures & Algorithms, Evaluation & Statistical Rigor, Computer Vision, OOP |

## Awards & Competitive Programming

- President's Scholarship of Distinction & Entrance Scholarship in Mathematics, University of Waterloo
- ICPC Local (Waterloo) — Rank 28 / ∼100 · 470+ LeetCode & 220+ DMOJ solved (1,497 points)
- Canadian Computing Competition — Junior: perfect score, Senior: top quartile · AIME Qualifier · COMC Distinction
- RoboCup International Junior Maze Rescue 2023 (Captain) — 2nd Place Supergroup, 17th Individual